

GROVE, OK, USA

WMO: 722092

Lat: 36.605N

Lon: 94.738W

Elev: 833

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53941

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.2	18.1	3.0	6.6	15.4	9.1	9.0	21.2	25.2	57.7	23.4	57.6	4.4	0	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	99.3	75.4	96.5	76.5	93.1	76.3	80.7	90.5	79.2	89.4	78.0	88.5	7.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	154.3	85.6	76.8	143.9	84.1	74.9	135.0	83.7	45.0	89.9	43.3	89.1	42.0	89.8	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	18.6	16.5	4.1	101.2	8.3	4.3	-1.9	104.3	-6.7	106.8	-11.4	109.3	-17.4	112.4		
			4.0	82.1	8.4	1.9	-2.0	83.5	-6.9	84.6	-11.6	85.6	-17.7	87.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.2	38.1	41.2	51.5	59.8	68.3	77.9	81.3	79.8	72.7	60.7	50.1	40.5	
	DBStd	17.50	10.88	11.44	10.87	8.44	7.92	5.19	5.10	5.83	6.62	9.16	9.60	9.60	
	HDD50	1259	392	290	110	16	2	0	0	0	0	15	113	321	
	HDD65	3622	836	668	437	197	60	1	0	0	13	195	455	760	
	CDD50	5003	22	44	158	310	569	836	970	925	680	347	115	27	
	CDD65	1889	1	2	19	41	162	386	505	460	243	61	8	1	
	CDH74	20211	2	15	140	388	1459	4120	5962	5162	2334	571	57	1	
	CDH80	9205	0	1	17	68	447	1806	3103	2646	959	152	6	0	
Wind	WSAvg	5.6	6.2	6.5	7.2	7.5	5.7	5.4	4.0	3.9	3.9	5.0	6.1	5.7	
Precipitation	PrecAvg	45.5	2.4	2.2	3.6	4.6	5.9	5.0	3.8	3.1	4.5	3.7	3.6	3.0	
	PrecMax	62.8	6.8	5.8	9.4	10.2	10.7	10.8	9.6	6.1	11.3	9.7	8.9	12.2	
	PrecMin	30.5	0.3	0.3	0.8	1.3	1.9	1.4	0.6	0.3	0.7	1.2	0.3	0.4	
	PrecStd	8.5	1.6	1.5	1.9	2.2	2.3	2.4	2.0	1.4	2.3	2.0	2.7	2.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.4	75.4	81.9	87.7	90.6	98.5	103.8	104.5	96.7	89.9	80.9	72.0	
		MCWB	59.0	62.4	67.3	67.8	72.9	76.5	73.6	75.2	73.8	72.6	65.0	63.0	
	2%	DB	66.2	71.7	78.9	81.5	88.4	94.7	99.4	99.6	91.2	84.0	74.6	66.3	
		MCWB	55.0	58.9	64.6	64.7	73.3	77.6	75.3	75.5	73.9	69.8	61.9	57.3	
	5%	DB	62.5	65.7	73.3	79.1	85.6	91.1	96.7	96.6	89.9	80.9	70.4	61.3	
		MCWB	51.8	57.6	62.1	64.2	71.7	76.2	76.5	76.4	73.5	66.5	61.1	53.4	
	10%	DB	56.5	61.1	70.2	75.0	81.6	89.9	93.1	92.5	86.4	76.7	66.5	56.6	
		MCWB	49.1	52.0	59.1	63.0	69.9	75.9	77.2	75.4	72.3	65.9	58.1	50.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.3	65.0	71.2	71.2	78.1	83.0	83.8	81.7	78.1	74.7	68.5	64.3
			MCDB	69.4	72.5	79.2	80.8	87.9	92.6	92.9	92.2	88.8	84.2	75.4	69.0
2%		WB	57.7	61.7	67.3	68.6	75.4	80.3	80.8	79.9	76.5	72.6	65.4	60.7	
		MCDB	62.7	68.9	75.0	77.5	84.0	89.8	90.1	90.8	87.3	81.0	71.6	64.6	
5%		WB	53.4	58.1	64.2	66.9	73.5	78.2	79.3	78.4	75.0	70.5	62.7	55.3	
		MCDB	60.2	64.6	72.5	75.6	81.0	88.1	89.2	89.4	84.9	76.7	68.6	59.3	
10%		WB	48.9	52.8	60.6	64.9	71.8	76.9	78.0	77.3	73.7	67.9	59.0	51.5	
		MCDB	57.1	59.4	68.2	72.7	79.8	86.7	88.5	88.3	82.8	73.8	65.1	55.3	
Mean Daily Temperature Range	5% DB	MDBR	20.7	20.5	21.9	21.9	20.1	19.5	21.0	20.9	21.9	22.4	22.2	19.7	
		MCDBR	29.4	27.8	26.2	25.3	22.9	21.6	24.7	25.3	24.8	25.9	26.1	26.6	
		MCWBR	18.8	18.3	15.1	13.4	10.8	8.5	7.2	7.5	9.3	12.5	16.9	18.1	
	5% WB	MCDBR	24.3	24.6	21.9	20.5	19.6	18.5	19.7	20.6	21.1	18.7	21.9	21.7	
		MCWBR	18.3	17.9	13.8	13.2	10.7	9.3	8.8	8.8	9.2	10.9	16.5	17.6	
Clear-Sky Solar Irradiance	taub		0.305	0.311	0.350	0.385	0.432	0.430	0.448	0.451	0.394	0.350	0.315	0.294	
	taud		2.465	2.462	2.390	2.309	2.236	2.269	2.264	2.234	2.358	2.462	2.505	2.515	
	Ebn at Noon		282	295	291	285	272	271	265	262	272	277	277	280	
	Edh at Noon		26	30	35	41	44	43	43	43	36	30	25	24	
All-Sky Solar Radiation	RadAvg		777	957	1275	1610	1794	2057	2049	1849	1569	1158	834	659	
	RadStd		69	141	75	95	154	178	126	126	119	146	93	82	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page